STEM Fellowship 2025-2026 High School Big Data Challenge

AI for Evaluating Irrigation Impact on Crop Yield in Northern Nigeria

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Abstract

There are many problems that affect the current state of Agriculture in Nigeria such as climate change and desertification, which contributes to the threat of food insecurity. The purpose of this project is to use climate indicators to identify the benefits and importance of surface irrigation in northern Nigeria. A data-driven method that uses historical weather data as well as agricultural and economic indicators from datasets from a variety of international organization such as the FAO, UN, and WFP among many others was used.

The methodology involves using a variety of data analysis techniques such as data visualization, data pre-processing and sorting, and machine learning models including linear regression and Random Forest Regression. Machine Learning models were refined to predict the most optimal environmental factors for cassava and corn yields. The use of data trees allows the ML model to identify several complex relationships between variables such as precipitation, temperature, and irrigation coverage, which has a much higher accuracy than simpler linear models.

This project mainly focuses on identifying meaningful patterns and trends that influence the yield of corn and cassava in Nigeria. It shows the potential for intelligent systems built on data-analysis as a means to identify the importance of climate indicators in plant growth and agricultural efficiency. The implications of these findings conclude the fact that ML and data science are very useful tools for addressing environmental challenges and for improving the stability of agriculture systems in the Sahel region by promoting the use of surface irrigation.

Keywords

- Machine Learning
- Sustainable Agriculture
- Food Insecurity
- Northern Nigeria
- Surface Irrigation

1 Introduction

As of August 2025, more than 30.6 million residents of Nigeria are projected to face acute food insecurity at Crisis level or worse [1]. As defined by the Integrated Food Security Phase Classification (IPC), IPC Phase 3 or "Crisis" means that families are facing acute malnutrition as well as prominent gaps in food consumption. During IPC Phase 3 families cannot easily obtain the nutrition they need and must resort to extreme measures such as selling off essential livelihood assets [2].

One source of food insecurity among the Nigerian population could be due to increasingly irregular and unpredictable crop yields caused by anomalous rainfall patterns created by climate change and global warming. These anomalous weather events lead to unpredictable floods and droughts, capable of ruining harvests and destroying crops. This is particularly dangerous because the 70% of the Nigerian population that uses some form of subsistence farming as a source of food and nutrition also rely on rainfall for the majority of their water [3]. Those who suffer the most from climate change related irregular rainfall patterns are rural farmers who are already lacking in resources. Without a way

to ensure a regular supply of water and/or increased flood protection, more and more Nigerians will risk facing acute food insecurity.

While statistics may show an increase in yield of the 2 main crops grown in Nigeria; cassava and corn, this increase is primarily due to the expansion of agricultural land rather than an actual increase in agricultural efficiency. Statistics demonstrate that despite increased yields, yield per hectare of certain crops, especially cassava, has been facing a marked decline [4].

One method that may both bridge the gaps between periods of rainfall reducing the risk of droughts and increase agricultural efficiency by stabilizing water supply on the yield of cassava and corn is irrigation. During preliminary research it was discovered that the majority of those who responded to the question of "What is the irrigation method used for [PLOT]?" 76.5% or 8617 cases responded with Surface Irrigation [5]. This makes sense as surface irrigation is one of the cheapest and easiest to implement, making it the ideal method for lower-income Nigerian farmers lacking in resources. Surface irrigation involves spreading water evenly through an entire field, this could be done by either flooding the area all at once or by using small channels and furrows.

While surface irrigation does seem like a relatively simple solution to the issue of erratic rainfall, how effective and sustainable is it really? By leveraging data analysis and machine learning techniques we hope to find the answer to the question of "Is surface irrigation an important factor to consistent and efficient cassava and corn yields in Nigeria?" We will focus on examining cassava and corn yield as a dependent variable while using various environmental factors such as rainfall and surface temperature as independent variables in order to determine the importance of a regular water supply to cassava and corn yields.

2 Materials & Methods

2.1 Materials

To ensure the data used is mostly unbiased and credible, an effort was made to only source data from trusted and reliable international organizations. The first of which was from World-Bank Climate Knowledge Portal [6], this dataset features extensive global environmental data including various climate indicators, as well as options to focus on specific countries. This was particularly important for our research as we aimed to limit our region of study to Nigeria, this dataframe allowed us to effectively isolate climate data without any further data processing. However, unfortunately, much of the timeframes available either did not align with our other datasets or were too general for our liking (e.g. yearly averages for precipitation). The dataset that we obtained from this source was Average Yearly Surface Temperatures in Nigeria from 1961-2023, for which we used to analyze the environmental impact of cassava and corn yields in both linear regression and random forest models.

Another major organization that data was extracted from was the Food and Agriculture Organization of the United Nations (FAO) [7], which provided peer-reviewed datasets focusing on crops and agriculture, particularly of the various seasonal yields of cassava and corn in Nigeria. This dataset proved to be incredibly helpful as it not only provided information on amount harvested but also yield in kg/ha which allowed us to more effectively analyze agricultural efficiency. This dataset was used as the independent variable in almost all of our models.

The third dataset used is Nigeria: Rainfall Indicators at Subnational Level [8] sourced from the Humanitarian Data Exchange, and created by the World Food Programme as a part of the WFP - Rainfall Indicators at a Subnational Level data series. This dataset was incredibly detailed and featured data taken from all 36 states of Nigeria as well as the Federal Capital Territory. The specific type of data contained was CHIRPS-GEFS short-term rainfall forecast data, the indicators used for the majority of our data analysis was 10 day rainfall [mm] (rfh) and 1 month rolling aggregation (r1h). This dataset was used to visualize weather patterns as well as to calculate environmental impacts on yield using linear regression and random forest models.

2.2 Data Processing

Due to the large amount of datapoints present in certain datasets, information had to be sorted and unnecessary data was filtered out. The bulk of this process was performed in Google Sheets where various cleaning tools such as functions and sorting by date allowed us to more efficiently standardize and increase the readability of our datasets. For example, the majority of areas measured for the rainfall indicators dataset had miniscule populations and a vastly smaller amount of input points, making the data more likely to be inaccurate. These data points were filtered out to ensure only data from the 36 states and the Federal Capital Territory remained. Area names were also added alongside the PCODEs already present in the dataset to

simplify the navigation and sorting processes. Finally, data was averaged across the 36 states and the FCT to create a Nigerian national average for the rfh and r1h indicators.

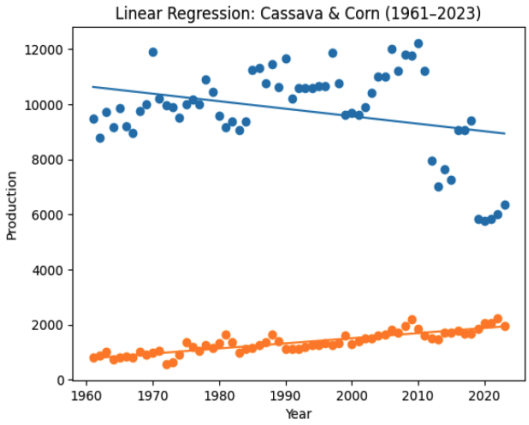
As another example, for the FAOSTAT cassava and corn yield data, yield in kg/ha was separated out and put onto a separate Google Sheet so that we could more easily use it for analysis. Many of our datasets were already complete with little or no missing values, when performing data pre-processing, sorting for relevant data points was much more important than cleaning. Any specific sorting for analysis or visualization (such as sorting for only the years of 2021-2025) was performed in Python alongside the code for visualization/machine learning models.

2.3 Data Analysis

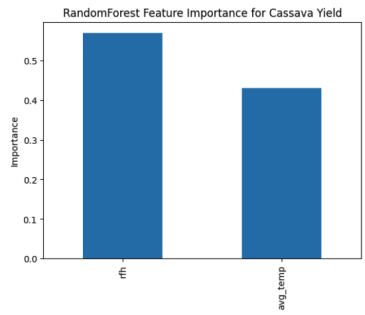
Various methods were used to turn cleaned datasets into actual interpretable results. This involved a variety of techniques ranging from basic data visualization using python to machine learning models such as linear regression and random forest. These models helped to highlight trends between a variety of variables as well as to predict crop yields under different conditions. Prior to performing data analysis, datasets underwent heavy pre-processing to isolate relevant variables and remove unnecessary or unusable information. Pre-processing during the coding stage allowed us to align timeframes for each dataset, ensuring time periods measured were accurate and then merging datasets in preparation for data analysis. Prior to model building, data visualization was performed to better understand already present climate and agricultural trends within Nigeria. Scatter plots were created to visualize yield harvested as kg/ha between the years of 1961 to 2023. Line graphs were made to visualize national averages for the CHIRPS-GEFS rainfall indicators as well as to compare rainfall in different areas of Nigeria. Finally, linear regression models were used to create simple predictions and analyze basic relationships between environmental variables and crop yield. For example annual mean surface temperature vs cassava and corn yields. Last but not least, random forest regression was used to predict and compare relationships between multiple environmental factors such as surface temperature and rainfall over time with cassava and corn yield. The random forest model was then refined and the optimal test size, random state, max depth, and n-estimators were determined by loop different combinations of parameters. Using the random forest model we were then able to determine the feature of highest importance as well as the partial dependency of each feature.

3 Results

By using linear regression and data visualization, we discovered clear trends in the relationship between individual climatic variables and the agricultural outcomes it influences. The long-term production trends for these crops are shown in Table 1.0, which displays the linear regression for cassava and corn from 1961 to 2023. Multiple other graphs were created such as line graphs demonstrating rainfall patterns over the years of 2021-2025.



The random forest model allowed for the observation of complex and non-linear relationships between multiple variables simultaneously, such as with variables like precipitation and temperature. By looping different combinations of parameters we determined that the optimal parameters for cassava yield was; test size - 0.6, random state - 80, n-estimators - 200, and max depth at 10. Although the R2 score was subpar at 0.1919, we were still able to determine a strong difference in feature importance between the two x-values with rfh having around 55 percent importance while average surface temperature had around 45 percent importance. This is represented in Table



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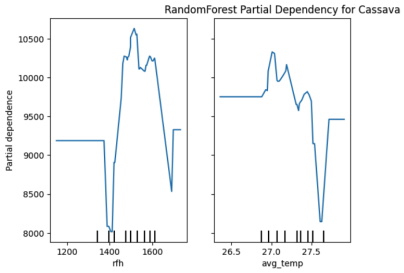
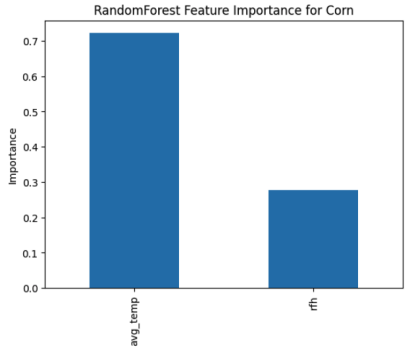
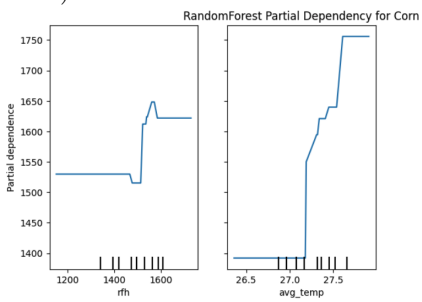


A strong correlation was also found between the amount of rainfall and surface temperature and cassava yield separately. Cassava yield is highest when rainfall is around 1500mm and average temperature is around 27.0 degrees celsius. (Table.3.0)





For corn yield the ideal parameters were determined to be: test size - 0.9, random state - 30, n-estimators - 200, and max depth at 30. Accuracy score (R2 score) saw a slight improvement compared to cassava at 0.2734. Temperature was found to have a much stronger importance in determining corn yield compared to rainfall with average temperature importance at above 70 percent while rainfall importance remained just below 30 percent. (Table 4.0). Corn yield is highest at temperatures above 27.25 degrees celsius and rainfall at around 1600mm. (Table 5.0)

4 Discussion

4.1 Analysis of Results

The primary aim of this study was to determine the importance of surface irrigation as a factor in improving corn and cassava yield. Our results demonstrate that cassava yield has a much higher reliance on water level compared to corn, while corn is much more dependent on temperature. Specifically our model shows that cassava may benefit much more from the regular and consistent water source provided by surface irrigation. However, our results also demonstrate that too much (above 1700mm per year) or too little (below 1400mm per year) water may drastically decrease cassava yields. This further highlights the direct correlation between irrigation as a consistent water source and the efficiency of cassava yield. Although rainfall is not of as much importance to corn yield compared to cassava, results show corn does still indeed benefit from rainfall levels above 1600mm.

4.2 Sources of Error

Many sources of error could have occurred during data processing, for one, a significant part of the rainfall data was filtered out due to having a small number of input points (n-pixels), this section of the data could have had a noticeable impact on the national averages for each of the rainfall indicators. Merging datasets and averaging a large amount of numbers could have also impacted the accuracy of final results. Furthermore, although the data sources used are highly credible organizations, there may still be potential bias in certain datasets, especially without local datasets to compare with. An example of this bias is the indicators that weren't measured in the datasets that were deemed irrelevant for the purpose of the original dataset, but that excludes potentially key information. In the future, a solution for this is to branch into more local data about different climate indicators, although datasets local to Nigeria aren't as commonly found. In that case, looking for institutionalized datasets from local governments may also be helpful.

4.3 Limitations

A major limiting factor in our analysis may be the broadness of our data. For example, crop data and temperature data was limited to yearly averages while rainfall data could be separated by the date, month, and year. FAOSTAT data lacked details about harvest seasons and growth cycles. This prevented more detailed analysis in regards to different timeframes as well as limited the types and amount of graphs we could make for visualization. Another limitation may be the geographical data (or lack thereof) regarding crop yield and temperature. This forced us to average rainfall data across regions and generalize a large range of areas into a single data point.

Only two crops were analyzed for this study; although cassava and corn are the two most popular agricultural plants in all of Nigeria, only focusing on them does not provide a full picture of Nigeria's agricultural landscape. As such, it may not demonstrate the full variety of issues different farmers specializing in the production of different crops would face. Besides the limitation in the amount of crops studied, there is also a limitation in the number of environmental factors studied. Crop growth relies on numerous different environmental factors, not just temperature and rain, a major limitation is found in the fact that we could not procure datasets for a greater variety of environmental indicators.

4.4 Expanding our Research

Our research may be expanded in a variety of ways including gathering more datasets on environmental and climate indicators, crop details and water usage, as well as soil conditions. More advanced machine learning models could be used to analyze a greater number of environmental factors, fine tuning feature importance and partial dependency results. A variety of methods of irrigation could be analyzed to see which one is most efficient for the needs of different crops.

5 Conclusion

This study investigated the impact of rainfall levels and surface temperature on cassava and corn yields in Nigeria. Our study demonstrated a correlation between the two environmental factors and the yield efficiency of cassava and corn crops, with rainfall being a major factor in the health of cassava crops while temperature was the major factor for corn crops. Both crops were found to benefit from a certain range of water levels. Especially in the case of cassava, too much or too little water may drastically reduce yield efficiency. Thus we are able to conclude that surface irrigation will to some level benefit both plants due to its ability to provide a stable and predictable water source. Due to limitations in certain datasets such as lack of harvest regions, lack of more specific timeframes, as well as data on a variety of other climate indicators; more precise and definitive conclusions could not be provided by this study.

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